

Assignment 6: Microeconometrics

Individual-Level Decisions, Risk, and Policy Targeting

Phase 6: Microeconometrics

Microeconometrics studies individual-level data where each observation represents an individual, household, firm, borrower, worker, patient, student, or transaction. This assignment focuses on practical econometric methods used when outcomes are binary, count-based, time-dependent, treatment-based, or affected by selection bias.

Problem Statement

A financial-inclusion program provides small loans, financial-literacy training, repayment reminders, and support services to low-income borrowers.

The research team wants to answer the following questions:

1. Which borrowers are most likely to default?
2. What factors are associated with repayment success?
3. How many missed payments should be expected?
4. How long does it take before a borrower defaults or exits the program?
5. Does financial-literacy training improve repayment outcomes?
6. Are treatment-effect estimates biased because some borrowers are more likely to receive training?
7. Are results affected by sample-selection bias?

Your task is to build a microeconomic analysis pipeline using individual-level borrower data.

Learning Objectives

By the end of this assignment, you should be able to:

- estimate and interpret binary choice models;
- compare logit and probit models;
- interpret odds ratios, marginal effects, and predicted probabilities;
- estimate count models using Poisson and negative binomial regression;
- detect and explain overdispersion;
- conduct basic survival analysis;
- estimate treatment effects using observational data;
- understand treatment selection and sample-selection bias;
- produce a clean empirical report with meaningful interpretation.

Dataset

You may use either a real dataset or a simulated dataset.

If you simulate the data, it must be realistic. Do not generate unrelated random variables. The relationships between variables should make economic sense.

The dataset should contain at least **3,000 observations**, where each row represents one borrower.

Suggested Variables

Borrower Characteristics

- age
- gender
- education years
- marital status
- household size
- urban/rural location
- monthly income
- savings balance

- prior credit history
- digital literacy score
- financial literacy score

Loan and Program Variables

- loan amount
- interest rate
- loan duration
- group lending status
- financial-literacy training
- repayment reminders
- program participation
- loan approval

Outcome Variables

- default status
- repayment success
- missed payment count
- months until default or exit
- program exit status

Part A: Data Preparation

A1. Create or Obtain the Dataset

Use either:

1. a real individual-level dataset, or
2. a simulated borrower-level dataset.

If you simulate the data, impose realistic relationships. For example:

- higher income should reduce default probability;
- stronger credit history should reduce default probability;
- larger loans may increase default risk;
- financial-literacy training may reduce missed payments;
- lower digital literacy may reduce program participation;
- treatment assignment may be partly targeted rather than fully random.

A2. Explain the Data-Generating Logic

Briefly explain:

- which variables affect default;
- which variables affect missed payments;
- which variables affect program participation;
- which variables affect treatment assignment;
- where selection bias may enter.

Part B: Exploratory Data Analysis

B1. Summary Statistics

Report summary statistics for all major variables:

- mean;
- standard deviation;
- minimum;
- maximum;
- number of observations;
- missing values.

B2. Group Comparisons

Compare borrowers by:

- default status;
- treatment status;
- loan approval status;
- program participation status.

Focus on variables such as income, education, loan amount, credit history, financial literacy, and savings balance.

B3. Visualizations

Create at least five useful plots. Suggested plots:

- distribution of monthly income;
- default rate by income group;
- missed payments by financial-literacy score;
- program participation by digital-literacy score;
- survival curve for time until default or exit;
- loan approval rate by credit-history group.

Part C: Binary Choice Models

Use binary outcomes such as:

- default status;
- loan approval;
- program participation;
- repayment success.

C1. Linear Probability Model

Estimate a linear probability model for default:

$$Default_i = \beta_0 + \beta_1 Income_i + \beta_2 Education_i + \beta_3 LoanAmount_i + \beta_4 CreditHistory_i + \beta_5 FinancialLiteracy_i$$

Discuss:

- coefficient signs;
- statistical significance;
- whether the results make economic sense;
- limitations of the linear probability model;
- whether any predicted probabilities fall outside the $[0, 1]$ range.

C2. Logit Model

Estimate a logit model for default.

Report and interpret:

- coefficients;
- odds ratios;
- predicted probabilities;
- marginal effects.

Explain clearly that logit coefficients are effects on log-odds, not direct changes in probability.

C3. Probit Model

Estimate a probit model using the same dependent variable and covariates.

Compare the probit model with the logit model:

- Are coefficient signs similar?
- Are significant variables similar?
- Are predicted probabilities similar?
- Which model is easier to interpret?

C4. Model Evaluation

Evaluate the binary models using:

- confusion matrix;
- accuracy;
- precision;
- recall;
- F1-score;
- ROC curve;
- AUC.

Explain why accuracy alone may be misleading if defaults are rare.

Part D: Count Models

Use missed payment count as the dependent variable.

D1. Poisson Regression

Estimate a Poisson regression model:

$$\text{MissedPayments}_i = f(\text{Income}_i, \text{Education}_i, \text{LoanAmount}_i, \text{CreditHistory}_i, \text{FinancialLiteracy}_i, \text{Tra} \dots)$$

Explain why Poisson regression is more appropriate than OLS for count data.

D2. Overdispersion Check

Compare the mean and variance of missed payment count.

If:

$$\text{Var}(Y) > E(Y)$$

then the data may be overdispersed.

Explain why overdispersion is a problem for Poisson regression.

D3. Negative Binomial Regression

Estimate a negative binomial regression model.

Compare it with the Poisson model in terms of:

- coefficient signs;
- standard errors;
- model fit;
- handling of overdispersion.

D4. Interpretation

Translate at least three count-model coefficients into plain language.

For example:

A one-unit increase in financial-literacy score is associated with fewer expected missed payments, holding other variables constant.

Part E: Survival Analysis

Use months until default or program exit as the time variable.

Use default or exit status as the event indicator.

E1. Kaplan-Meier Survival Curve

Estimate and plot a Kaplan-Meier survival curve.

Explain:

- what survival probability means;
- how survival changes over time;
- whether default risk appears early or late in the loan period.

E2. Group Comparison

Plot separate survival curves for at least two groups, such as:

- trained versus untrained borrowers;
- high versus low income borrowers;
- good versus poor credit-history borrowers;
- group lending versus individual lending.

Discuss whether survival differs meaningfully across groups.

E3. Cox Proportional Hazards Model

Estimate a Cox proportional hazards model.

Suggested covariates:

- monthly income;
- loan amount;
- financial-literacy score;
- prior credit history;
- training status;
- group lending;
- repayment reminders.

Interpret hazard ratios.

A hazard ratio below 1 indicates lower event risk at a given point in time. A hazard ratio above 1 indicates higher event risk.

Part F: Treatment Effects

Use financial-literacy training or intervention assignment as the treatment variable.

Possible outcomes:

- default status;
- repayment success;
- missed payment count;
- months until default or exit.

F1. Naive Difference in Means

Compare the average outcome between treated and untreated borrowers.

Discuss why this difference may not be causal.

F2. Regression Adjustment

Estimate a regression model controlling for observed borrower characteristics:

$$Default_i = \beta_0 + \beta_1 Training_i + \beta_2 Income_i + \beta_3 Education_i + \beta_4 CreditHistory_i + \beta_5 LoanAmount_i + \varepsilon_i$$

Interpret the treatment coefficient.

Discuss whether the estimated treatment effect changes after adding controls.

F3. Propensity Score Model

Estimate the probability of receiving treatment using logit or probit:

$$P(Training_i = 1) = f(Income_i, Education_i, CreditHistory_i, DigitalLiteracy_i, Urban_i, Age_i)$$

Save the predicted propensity score.

Explain which borrowers were more likely to receive treatment.

F4. Matching or Weighting

Use either:

1. propensity score matching, or
2. inverse probability weighting.

Compare:

- naive difference in means;
- regression-adjusted estimate;
- matching or weighting estimate.

Discuss whether the estimated treatment effect is stable.

F5. Heterogeneous Treatment Effects

Test whether the intervention works differently for different groups.

Use interaction terms such as:

$$Training_i \times LowIncome_i$$

or

$$Training_i \times LowFinancialLiteracy_i$$

Discuss whether some borrower groups benefit more than others.

Part G: Sample Selection

Microeconomic data often suffers from selection bias.

For example, default is observed only for borrowers whose loans were approved. If loan approval depends on creditworthiness, the observed default sample is not random.

G1. Explain the Selection Problem

Identify one selection issue in your dataset.

Explain why ignoring this selection problem may bias the results.

G2. Selection Equation

Estimate a probit model for loan approval:

$$\text{LoanApproved}_i = f(\text{Age}_i, \text{Income}_i, \text{Education}_i, \text{CreditHistory}_i, \text{DigitalLiteracy}_i, \text{Urban}_i, \text{Savings}_i)$$

This is the selection equation.

G3. Outcome Equation

Estimate the default or repayment equation only among approved borrowers:

$$\text{Default}_i = f(\text{Income}_i, \text{Education}_i, \text{LoanAmount}_i, \text{CreditHistory}_i, \text{FinancialLiteracy}_i, \text{Training}_i)$$

Discuss why this may be biased if selection is ignored.

G4. Heckman-Style Correction

If possible, implement a Heckman-style two-step correction:

1. Estimate the selection equation using probit.
2. Compute the inverse Mills ratio.
3. Include the inverse Mills ratio in the outcome equation.

Explain conceptually what the inverse Mills ratio is correcting for.

G5. Compare Results

Compare the outcome model:

1. without selection correction;

2. with selection correction.

Discuss whether:

- coefficients changed;
- the selection correction term is significant;
- selection bias appears important.

Part H: Final Interpretation

Write a final empirical interpretation answering:

1. Which borrowers are most likely to default?
2. Which borrowers miss the most payments?
3. Which factors improve repayment behavior?
4. Does financial-literacy training appear effective?
5. Is the estimated treatment effect likely causal or only associational?
6. Are treatment effects heterogeneous?
7. Is sample-selection bias important?
8. Which model is most useful for prediction?
9. Which model is most useful for interpretation?
10. What policy recommendation would you give?

Required Submission

Submit the following:

1. A clean Jupyter notebook containing data preparation, EDA, models, plots, and interpretation.
2. A short empirical report of 2,000–3,000 words.
3. A separate Python file containing reusable functions.
4. A README explaining the project, dataset, methods, and main findings.

Suggested Python Libraries

You may use:

- pandas
- numpy
- matplotlib
- seaborn

- statsmodels
- scikit-learn
- lifelines
- scipy

Optional libraries:

- linearmodels
- econml
- dowhy
- causalm1

Use optional causal libraries only if you understand the underlying method.

Minimum Requirements

Your assignment must include at least:

1. one linear probability model;
2. one logit model;
3. one probit model;
4. one Poisson model;
5. one negative binomial model;
6. one Kaplan-Meier survival curve;
7. one Cox proportional hazards model;
8. one treatment-effect estimate using regression adjustment;
9. one propensity-score-based estimate;
10. one sample-selection discussion or correction.

Suggested Folder Structure

```
assignment_6_microeconometrics/  
|  
|-- data/  
|   |-- micro_dataset.csv  
|  
|-- notebooks/  
|   |-- assignment_6_microeconometrics.ipynb  
|  
|-- src/  
|   |-- data_generation.py  
|   |-- plots.py  
|   |-- models.py  
|   |-- treatment_effects.py  
|  
|-- outputs/  
|   |-- figures/  
|   |-- tables/  
|  
|-- report/  
|   |-- assignment_6_report.pdf  
|  
|-- README.md  
|-- requirements.txt
```

Grading Rubric

Component	Weight
Data preparation and understanding	10%
Exploratory data analysis	10%
Binary choice models	15%
Count models	10%
Survival analysis	15%
Treatment effects	15%
Sample selection analysis	10%
Interpretation and policy discussion	10%
Code quality and reproducibility	5%

Restrictions

- Do not submit a notebook that only runs models without interpretation.
- Do not claim causality from a simple regression unless the assumptions are clearly stated.
- Do not ignore treatment selection or sample-selection bias.
- Do not use black-box causal libraries without explaining the method.
- Do not report coefficients without explaining their real-world meaning.

Final Deliverable

Your final submission should answer the following question:

Using individual-level borrower data, what can microeconomic methods tell us about default risk, repayment behavior, treatment effectiveness, and selection bias in a financial-inclusion program?

The assignment should read like a small applied microeconomics paper, not just a coding exercise.